SOLUTIONS: CONTROL SYSTEMS – 2020/2021

1. Solution

a) Consider the two integrator blocks shown in the block scheme in Figure 1.1 in the text of the exam paper with outputs $v(t)$ and $w(t)$, respectively. Clearly only these blocks represent dynamic systems. Therefore, a *possible choice* for the state variables is:

$$x_1(t) := v(t); \quad x_2(t) := w(t)$$

Hence, we define the state and output vectors:

$$x(t) := \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}; \quad y(t) := \begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix}$$

Accordingly, by inspecting the block scheme in Figure 1.1, we obtain:

$$\dot{x}_1(t) = r(t); \quad \dot{x}_2(t) = v(t)$$

Moreover:

$$\begin{cases} r(t) = u(t) - 6v(t) - 6w(t) \\ y_1(t) = Kw(t) + v(t) \\ y_2(t) = w(t) \end{cases}$$

After some algebra, we get:

$$\begin{cases} \dot{x}_1(t) = -6x_1(t) - 6x_2(t) + u(t) \\ \dot{x}_2(t) = x_1(t) \\ y_1(t) = x_1(t) + Kx_2(t) \\ y_2(t) = x_2(t) \end{cases}$$

Accordingly, the matrices A, B, C, D of the state space description are given by:

$$A = \begin{bmatrix} -6 & -6 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 1 & K \\ 0 & 1 \end{bmatrix}, D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

[12 Marks]

b) The stability properties of the dynamic system shown in Figure 1.1 as a function of the scalar $K \in \mathbb{R}$ only depend on the state matrix A of any state-space description of the system. From the answer to Question 1-a), we have:

$$A = \begin{bmatrix} -6 & -6 \\ 1 & 0 \end{bmatrix}$$

which **does not depend on** K. Therefore, the stability properties also do not depend on K.

The characteristic polynomial is given by:

$$p_A(s) = \det(sI - A) = \det \begin{bmatrix} s+6 & 6 \\ -1 & s \end{bmatrix} = s \cdot (s+6) + 6 = s^2 + 6s + 6$$

Hence, the eigenvalues of A are:

$$\lambda_1 = -3 - \sqrt{3}; \quad \lambda_1 = -3 + \sqrt{3}$$

Answers

Since $\lambda_1 < 0$ and $\lambda_2 < 0$, we conclude that the dynamic system is asymptotically stable $\forall K \in \mathbb{R}$.

[8 Marks]

c) By inspecting the state equations

$$\begin{cases} \dot{x}_1(t) &= -x_1(t) + u(t) \\ \dot{x}_2(t) &= 0.5x_2(t) \\ \dot{x}_3(t) &= 0.2x_1(t) - 0.2x_3(t) \\ y(t) &= x_1(t) + x_2(t) \end{cases}$$

one immediately gets matrices A, B, C, D:

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 1/5 & 0 & -1/5 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}, D = 0$$

By noticing that matrix A is lower-triangular, the characteristic polynomial can be easily calculated

$$p_A(s) = \det(sI - A) = \det \begin{bmatrix} s+1 & 0 & 0 \\ 0 & s-1/2 & 0 \\ -1/5 & 0 & s+1/5 \end{bmatrix}$$

which, after some algebra, gives

$$p_A(s) = (s+1) \cdot (s-1/2) \cdot (s+1/5) = \cdots = s^3 + \frac{7}{10}s^2 - \frac{2}{5}s - \frac{1}{10}$$

Since $x(0) = [0, 0, 0]^\top$, we have:

$$y(t) = \mathscr{L}^{-1}\left[C \cdot (sI - A)^{-1} \cdot B \cdot U(s)\right]$$

where $U(s) = \mathscr{L}[u(t)]$ and $\mathscr{L}[\cdot]$ denotes the Laplace transform operator.

Thus:

$$C \cdot (sI - A)^{-1} \cdot B = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} s+1 & 0 & 0 \\ 0 & s-1/2 & 0 \\ -1/5 & 0 & s+1/5 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Easy calculations yield:

$$\begin{bmatrix} s+1 & 0 & 0 \\ 0 & s-1/2 & 0 \\ -1/5 & 0 & s+1/5 \end{bmatrix}^{-1} = \cdots = \begin{bmatrix} \frac{1}{s+1} & 0 & 0 \\ 0 & \frac{2}{2s-1} & 0 \\ \frac{1}{(5s+1)(s+1)} & 0 & \frac{5}{5s+1} \end{bmatrix}$$

Therefore:

$$C \cdot (sI - A)^{-1} \cdot B = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{s+1} & 0 & 0 \\ 0 & \frac{2}{2s-1} & 0 \\ \frac{1}{(5s+1)(s+1)} & 0 & \frac{5}{5s+1} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{s+1}$$

Answers

Hint: by inspecting matrices B and C, it follows immediately that there is no need to compute all elements of matrix $(sI-A)^{-1}$; instead, only two elements of $(sI-A)^{-1}$ are needed: the element in row 1, column 1, and the element in row 2, column 1.

Clearly, the input $u(t) = 1(t-1)$ is a unit step-function delayed by 1 sec, and hence:

$$U(s) = \mathscr{L}[u(t)] = \mathscr{L}[1(t-1)] = \frac{e^{-s}}{s}$$

Since the system is time-invariant, it is convenient to first determine the output function $\widetilde{y}(t)$ generated by the non-delayed input $\widetilde{u}(t) = 1(t)$ and apply the delay afterwards: $y(t) = \widetilde{y}(t-1)$.

Specifically, we first compute:

$$\widetilde{y}(t) = \mathscr{L}^{-1}[\widetilde{Y}(s)] = \mathscr{L}^{-1}\left[C \cdot (sI-A)^{-1} \cdot B \cdot \widetilde{U}(s)\right]$$

where

$$\widetilde{U}(s) = \mathscr{L}[1(t)] = \frac{1}{s}; \quad \widetilde{Y}(s) = \mathscr{L}[\widetilde{y}(t)]$$

Hence:

$$\widetilde{Y}(s) = \frac{1}{s(s+1)}$$

The partial-fraction expansion gives:

$$\frac{1}{s(s+1)} = \frac{\alpha}{s} + \frac{\beta}{s+1}$$

where:

$$\alpha = \left.\frac{1}{s+1}\right|_{s=0} = 1; \quad \beta = \left.\frac{1}{s}\right|_{s=-1} = -1$$

Thus, we get:

$$\widetilde{y}(t) = \mathscr{L}^{-1}[\widetilde{Y}(s)] = \mathscr{L}^{-1}\left[\frac{1}{s} - \frac{1}{s+1}\right] = \left[1 - e^{-t}\right] \cdot 1(t)$$

and finally:

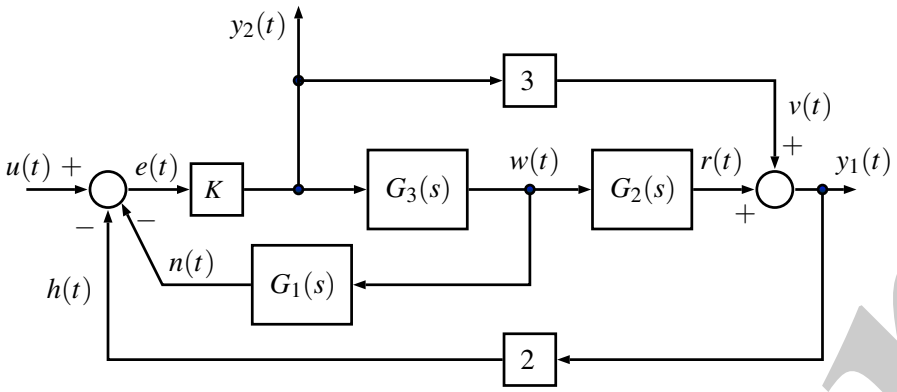
$$y(t) = \widetilde{y}(t-1) = \left[1 - e^{-(t-1)}\right] \cdot 1(t-1)$$

[20 Marks]

Answers

2. Solution

a) Referring to the block scheme shown in Figure 2.1 in the text of the exam paper, it is convenient to introduce the auxiliary variables $e(t)$, $h(t)$, $n(t)$, $w(t)$, $r(t)$, and $v(t)$, as shown in figure below:



By inspecting the above block scheme, the following equations are obtained (we drop the dependence on t for simplicity):

$$\begin{cases} y_2 = K \cdot e \\ e = u - h - n \\ n = G_1(s) \cdot w \\ h = 2 \cdot y_1 \\ w = G_3(s) \cdot y_2 \\ r = G_2(s) \cdot w \\ y_1 = r + v \\ v = 3 \cdot y_2 \end{cases}$$

By eliminating all auxiliary variables, we obtain:

$$y_2 = K \cdot [u - 6y_2 - 2 \cdot G_2(s) \cdot G_3(s) \cdot y_2 - G_1(s) \cdot G_3(s) \cdot y_2]$$

thus getting

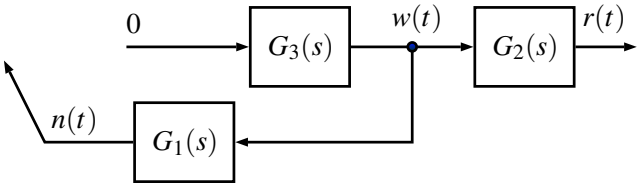
$$y_2 = \frac{K}{1 + 6 \cdot K + K \cdot G_3(s)[G_1(s) + 2 \cdot G_2(s)]} \cdot u$$

and hence

$$G_{u,y_2}(s) = \frac{K}{1 + 6 \cdot K + K \cdot G_3(s)[G_1(s) + 2 \cdot G_2(s)]}$$

[8 Marks]

b) For $K = 0$, the block scheme shown in the figure in the above answer to Question 2-a) can be reduced to the scheme in figure below:



The three blocks with transfer functions $G_1(s)$, $G_2(s)$, and $G_3(s)$ involve only series-interconnections. Also, recall that the transfer functions $G_1(s)$, $G_2(s)$,

Answers

and $G_3(s)$ have no hidden common factors in their numerators and denominators.

Hence, for each $i = 1,2,3$ the set of eigenvalues of the state matrix A_i of any state-space description of the dynamic system represented by the transfer function $G_i(s)$ is included in the set of eigenvalues of the state matrix A of any state-space description of the entire dynamic system. Therefore, we can conclude that the instability of either $G_1(s)$ or $G_2(s)$ or $G_3(s)$ implies the instability of the whole dynamic system shown in Figure 2.1 in the text of the exam paper.

[6 Marks]

c) For $K \neq 0$, the block scheme is the entire one shown in Figure 2.1 also reported in the answer to Question 2-a).

By inspecting this block scheme, it clearly turns out that all transfer functions $G_1(s)$, $G_2(s)$, and $G_3(s)$ show up in some feedback interconnection. Hence, in general, the eigenvalues of the state matrix A_i of any state-space description of the dynamic system represented by the transfer function $G_i(s)$ are different from the eigenvalues of the state matrix A of any state-space description of the entire dynamic system. Therefore, no conclusions can be drawn on the stability properties of the whole dynamic system shown in Figure 2.1 in the text of the exam paper from the instability of either $G_1(s)$ or $G_2(s)$ or $G_3(s)$.

[6 Marks]

Answers

3. Solution

a) We consider a controller $C(s)$ with the structure:

$$C(s) = C_1(s) \cdot C_2(s)$$

where $C_1(s)$ is designed to guarantee that the Static Requirement i) in Question 3-a) is satisfied while $C_2(s)$ obeys the constraint $C_2(0) = 1$ (so as not to influence the satisfaction of the static Requirement i)) and is devoted to ensure that the Dynamic Requirements ii), iii) and iv) are satisfied.

We follow a loop-shaping design procedure setting $C_2(s) = 1$ as a first attempt.

To meet the Static Requirement i) we set

$$C_1(s) = \frac{\mu_C}{s}$$

where $\mu_C \in \mathbb{R}, \mu_C > 0$. By now, there is no need to select a specific value of $\mu_C > 0$ to meet the Static Requirement i) due to the addition of the pole $s = 0$ by the controller $C_1(s)$.

To meet Requirement ii) with a reasonable margin, we **choose**

$$t_{s,0.01} = 9 \text{ sec}$$

Moreover, owing to Requirement iv) for which φ_m has to satisfy $\varphi_m \geq 40^\circ$, we consider a possible scenario where the closed-loop dominant poles are complex conjugate with natural angular frequency ω_n and damping ratio ξ.

Hence, using the relationship

$$\Delta\% = 100 \cdot e^{-\pi\xi/\sqrt{1-\xi^2}}$$

we obtain:

$$\Delta\% \leq 20\% \Longrightarrow 100 \cdot e^{-\pi\xi/\sqrt{1-\xi^2}} \leq 20 \Longrightarrow \xi^2 \geq \frac{(\ln 5)^2}{\pi^2 + (\ln 5)^2} \simeq 0.2$$

Again, to make a choice with a reasonable margin, we **choose**

$$\xi = 0.5$$

From the approximate formula (valid for second-order systems with complex conjugate poles)

$$t_{s,0.01} \simeq \frac{4.6}{\xi\,\omega_n}$$

and the above choice $t_{s,0.01} = 9$ sec, we obtain

$$\omega_n \simeq \frac{4.6}{9 \cdot 0.5} \simeq 1\,\text{rad/sec}$$

Now, we recall the standard dominant-poles approximation for the crossover angular frequency ω_c:

$$\omega_c \simeq \omega_n$$

Thus, by designing the controller $C(s)$ in such a way that

$$\omega_c \geq 1\,\text{rad/sec}$$

it turns out that Requirements ii) and iii) are achievable.

Answers

Furthermore, we recall the approximate relationship for the damping ratio ξ of the closed-loop dominant poles when $\omega_c \simeq \omega_n$:

$$\xi \simeq \sin\left(\frac{\varphi_m}{2}\right) \Longrightarrow \varphi_m \simeq 2 \cdot \arcsin(\xi)$$

Plugging in the chosen value of the damping ratio $\xi = 0.5$, we get

$$\varphi_m \simeq 60^\circ$$

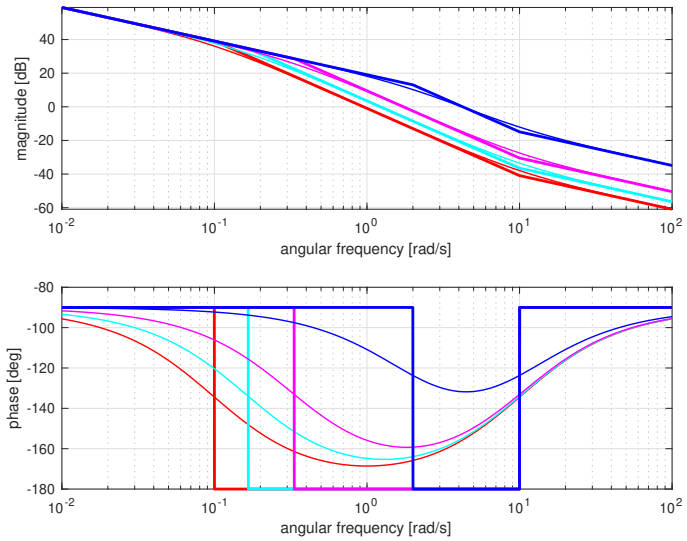
which would satisfy Requirement iv) with a very good margin.

The next step is to design the controller $C(s)$ by an iterative loop-shaping procedure aiming at achieving a crossover angular frequency $\omega_c \geq 1\,\text{rad/sec}$ and a phase margin $\varphi_m \simeq 60^\circ$ taking into account that the time-constant T in the controlled system's transfer function $G(s)$ is unknown and satisfies the constraint $T \in [0.5,\ 10]$.

To gain some useful insight for our loop-shaping procedure, we set $C(s) = C_1(s)$ with $\mu_C = 1$ thus obtaining the open-loop transfer function

$$L_1(s) = 9 \cdot \frac{1 + 0.1\,s}{s(1 + T\,s)}$$

and draw the Bode diagrams of $L_1(s)$ (both the asymptotic approximations and the true ones) in figure below for several admissible values of T: $T = 10$ (red colour), $T = 6$ (cyan colour), $T = 3$ (magenta colour), and $T = 0.5$ (blue lines).



By visually inspecting the Bode diagrams in figure above, it follows that the crossover frequency ω_c and the phase margin φ_m both increase for decreasing values of T.

Hence, we proceed in our loop-shaping controller design by keeping the structure of $C(s) = C_1(s)$ and by trying to select a suitable value of the gain μ_C such that Requirements i), ii), iii), and iv) are simultaneously satisfied in the case $T = 10$ that corresponds to the smallest values of ω_c and φ_m.

Again, by visually inspecting the Bode diagrams in figure above, we observe that crossover frequencies ω_c larger than $10\,\text{rad/sec}$ should allow getting values of the phase margin φ_m larger than 60°. Let us make three attempts $\omega_c^{(1)}, \omega_c^{(2)}, \omega_c^{(3)}$:

A-level

- Set $\omega_c^{(1)} = 10\,\text{rad/sec}$:

$$\arg L(j\omega_c^{(1)}) = \arg(1+j10/10)\cdot\frac{180^\circ}{\pi} - 90^\circ - \arg(1+j10\cdot 10)\cdot\frac{180^\circ}{\pi} \simeq -134.4^\circ$$

Hence, $\varphi_m^{(1)} = 180^\circ + \arg L(j\omega_c^{(1)}) \simeq 45.6^\circ \not\geq 60^\circ$ (Req. iv) not satisfied)

- Set $\omega_c^{(2)} = 15\,\text{rad/sec}$:

$$\arg L(j\omega_c^{(2)}) = \arg(1+j15/10)\cdot\frac{180^\circ}{\pi} - 90^\circ - \arg(1+j15\cdot 10)\cdot\frac{180^\circ}{\pi} \simeq -123.3^\circ$$

Hence, $\varphi_m^{(2)} = 180^\circ + \arg L(j\omega_c^{(2)}) \simeq 56.3^\circ \not\geq 60^\circ$ (Req. iv) not satisfied)

- Set $\omega_c^{(3)} = 20\,\text{rad/sec}$:

$$\arg L(j\omega_c^{(1)}) = \arg(1+j20/10)\cdot\frac{180^\circ}{\pi} - 90^\circ - \arg(1+j20\cdot 10)\cdot\frac{180^\circ}{\pi} \simeq -116.3^\circ$$

Hence, $\varphi_m^{(3)} = 180^\circ + \arg L(j\omega_c^{(3)}) \simeq 63.7^\circ > 60^\circ$ (Req. iv) satisfied)

Thus, a crossover frequency $\omega_c = \omega_c^{(3)} = 20\,\text{rad/sec}$ would be satisfactory in the worst-case $T = 10$ and hence for all $T \in [0.5, 10]$.

The last step to conclude the loop-shaping design procedure of $C(s)$ is to calculate the value of the gain μ_C yielding the desired crossover frequency $\omega_c^{(3)}$:

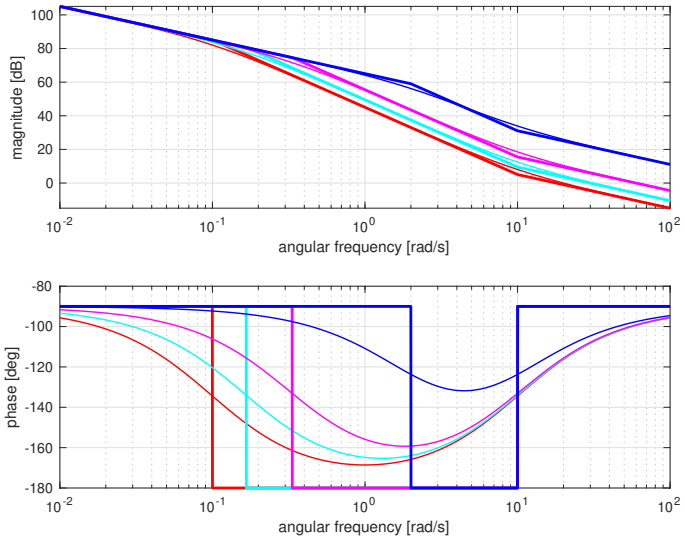
$$\left|L(j\omega_c^{(3)})\right| = 1 \Longrightarrow 9\mu_C\frac{|1+j20/10|}{|(j20)|\,|(1+j20\cdot 10)|} = 1 \Longrightarrow \mu_C \simeq 198.8$$

Hence, the design of the controller is completed by **choosing**:

$$C(s) = C_1(s) = \frac{200}{s}$$

thus getting the open-loop transfer function

$$L_2(s) = 1800\cdot\frac{1+0.1s}{s(1+Ts)}$$



In figure above, we draw the Bode diagrams of $L_2(s)$ (both the asymptotic approximations and the true ones) for the the same values of T as in figure above

Answers

(that is, $T = 10$ (red colour), $T = 6$ (cyan colour), $T = 3$ (magenta colour), and $T = 0.5$ (blue lines)). These diagrams clearly show that the design of $C(s)$ for the worst case scenario ensures that the Requirements i)-iv) are actually satisfied for all $T \in [0.5, 10]$.

[18 Marks]

b) Consider the block scheme shown in Figure 3.1 in the text of the exam paper. The closed-loop transfer functions $T_{d,y}(s)$ and $T_{n,y}(s)$ from the inputs $d(t)$ and $n(t)$ to the output $y(t)$, respectively, are given by

$$T_{d,y}(s) = \frac{Y(s)}{D(s)} = \frac{1}{1+L_2(s)}; \quad T_{n,y}(s) = \frac{Y(s)}{N(s)} = -\frac{L_2(s)}{1+L_2(s)}$$

where $L_2(s)$ denotes the open-loop transfer function

$$L_2(s) = 1800 \cdot \frac{1+0.1s}{s(1+Ts)}$$

defined in the answer to Question 3-a).

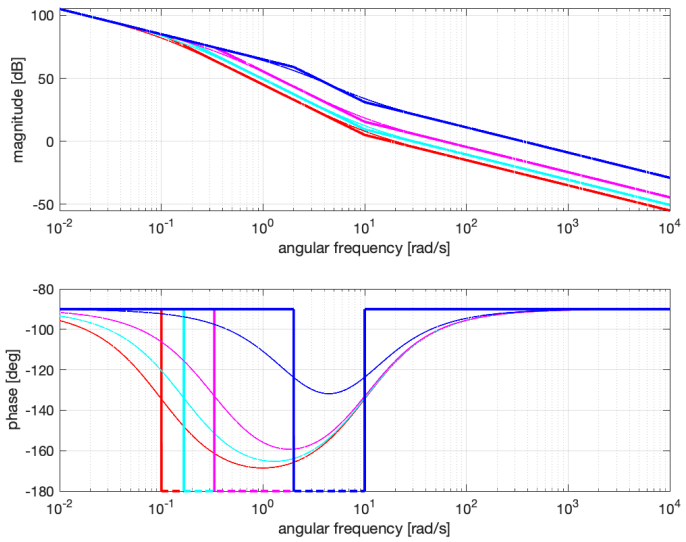
Denoting by $y_{d,\infty}(t)$ the regime (steady-state) output function when $w(t) = n(t) = 0, \forall t$ and by $y_{n,\infty}(t)$ the regime (steady-state) output function when $w(t) = d(t) = 0, \forall t$, we have:

$$y_\infty(t) = y_{d,\infty}(t) + y_{n,\infty}(t)$$

Since $d(t) = 5 \cdot 1(t)$ and the static Requirement i) is satisfied, $y_{d,\infty}(t) = 0, \forall t$ and hence, thanks to the frequency response theorem, it follows that:

$$y_\infty(t) = y_{n,\infty}(t) = 0.1 \cdot \left| \frac{L_2(j\bar{\omega})}{1+L_2(j\bar{\omega})} \right|$$

In figure below, we plot again the Bode diagrams of $L_2(s)$ given in the answer to Question 3-a) on a wider frequency range.



It is evident that at high-enough frequency the following approximation is acceptable:

$$\left| \frac{L_2(j\omega)}{1+L_2(j\omega)} \right| \simeq |L_2(j\omega)|$$

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and hence, according to this approximation, we have

$$|L_2(j\bar{\omega})|_{\text{dB}} < -20\text{dB} \Longrightarrow |y_\infty(t)| < 10^{-2} \qquad (\star)$$

Also, inspecting the above Bode diagrams, it is evident that the worst-case scenario to be considered now is the one for $T = 0.5$ (blue lines): guaranteeing the satisfaction of the above inequality $(\star)$ for $T = 0.5$ implies that the inequality $(\star)$ is satisfied for every $T \in [0.5, 10]$.

Hence, we consider

$$\widetilde{L}_2(s) = 1800 \cdot \frac{1+0.1s}{s(1+0.5s)}$$

and we look for a suitable angular frequency $\bar{\omega}$ such that

$$\left|\widetilde{L}_2(j\bar{\omega})\right|_{\text{dB}} < -20\text{dB}$$

The asymptotic approximation of the Bode diagram for $\left|\widetilde{L}_2(j\bar{\omega})\right|_{\text{dB}}$ is a piecewise linear function characterised by the following straight-line equations:

- $0 < \omega \leq 2$:
  $-20\log(\omega) + 20\log(1800) \simeq -20\log(\omega) + 65.1\text{dB}$
- $2 < \omega \leq 10$:
  $-40[\log(\omega) - \log(2)] - 20\log(2) + 20\log(1800) \simeq -40\log(\omega) + 71.1\text{dB}$
- $\omega \geq 10$:
  $-20[\log(\omega) - \log(10)] - 40[\log(10) - \log(2)] - 20\log(2) + 20\log(1800)$
  $\simeq -20\log(\omega) + 51.1\text{dB}$

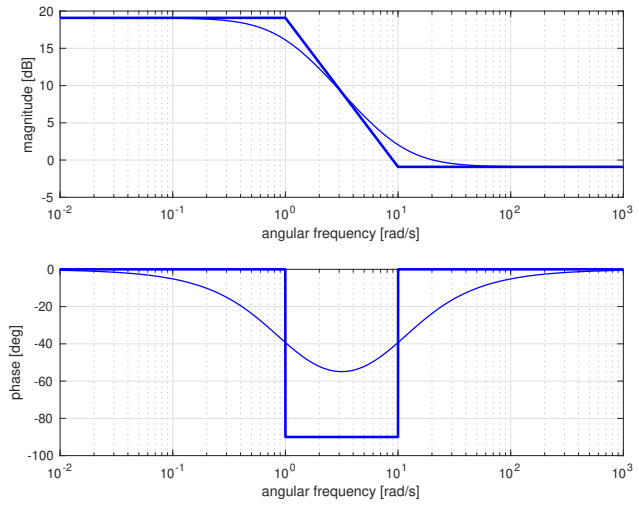
Thus:

$$-20\log(\bar{\omega}) + 51.1\text{dB} < -20\text{dB} \Longrightarrow \bar{\omega} > 3589.2\,\text{rad/sec}$$

[14 Marks]

c) We set $d(t) = 0, \forall t$, $n(t) = 0, \forall t$, $C(s) = 1$, and $T = 1$sec. Hence, the open-loop transfer function now reads:

$$L(s) = 9 \cdot \frac{1+0.1s}{1+s}$$



The Bode diagrams of the open-loop transfer function $L(s)$ are shown in the above figure where the asymptotic approximation of the Bode diagram of $|L(j\omega)|_{\text{dB}}$

Answers

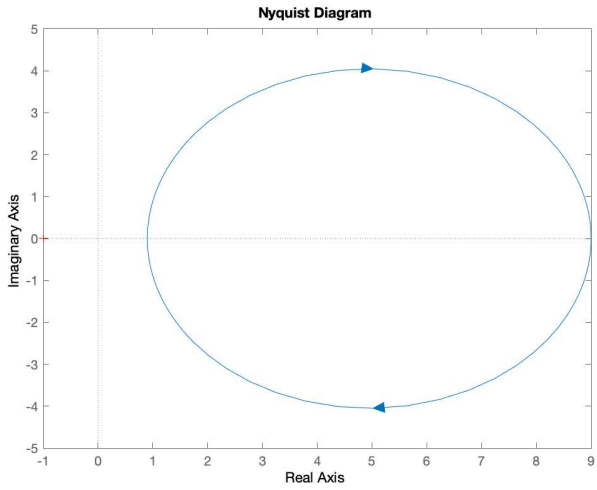
is a piecewise linear function characterised by the following straight-line equations:

- $0 < \omega \leq 1$:
  $20\log(9) \simeq 19.1\text{dB}$
- $1 < \omega \leq 10$:
  $-20[\log(\omega) - \log(1)] + 20\log(9) \simeq -20\log(\omega) + 19.1\text{dB}$
- $\omega \geq 10$:
  $-20\log(10) + 20\log(9) \simeq -0.91\text{dB}$

Hence, the main characteristics of the polar diagram of $L(j\omega)$ are:

- $L(j0) = 9$
- $\lim_{\omega \to +\infty} |L(j\omega)| = 0.9$
- $\arg L(j0) = 0^\circ$
- $\lim_{\omega \to +\infty} \arg L(j\omega) = 0$
- $\arg L(j\omega) \in (-90^\circ, 0^\circ]\,,\ \forall\, \omega \in [0, +\infty)$

The Nyquist diagram is drawn in figure below.



The number of rotations around the point $(-1, 0)$ is well-defined and hence the Nyquist closed-loop stability criterion can be used.

[8 Marks]

Answers